

Ecological and economic perspectives of two decades of change in a traditional fishing ground in the Philippines. *Supplementary material.*

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SUPPLEMENTARY TABLES

Table S1. Fishing gear types included in the Visayan Sea model grouped by commercial and municipal fishing sectors

	Fleet groups	Abbreviation/Code	Average fishers or crew per unit	Est. number of units, 1997	Est. number of units, 2018
	<u>Commercial (> 3 G.T.)</u>				
1	Danish seines	DS	13.5	489	87
2	Purse seines	PS	16.5	71	58
3	Ring nets	RN	13.5	44	116
4	Bag nets	BN	4.5	41	94
5	Otter trawls	OT	6.5	471	172
6	Mid-water trawls	MWT	9	112	40
7	Modified seine nets	MSN	5	69	92
	<u>Municipal (< 3 G.T.)</u>				
8	“Baby” Danish seines	BDS	5.5	288	236
9	“Baby” trawls	BOT	1.5	922	1797
10	Beach seines and variations	BeachS	6	94	645
11	Stationary lift nets and bag nets	SLift-BagN	1.5	93	293
12	Bottom-set gillnets and variations	BSGN	1.5	2573	13671
13	Crab nets	CrabN	1.5	1815	6764
14	Crab liftnets	CrabLN	1	706	693
15	Crab pots	CrabP	1	583	4305
16	Drift and surface gillnets	DGN	2	1736	4605
17	Drive-in nets and scare nets	DrGN	2.5	397	833
18	Encircling gillnets	EGN	3.5	641	743
19	Fish corrals, fyke nets, filter nets	FCor	1.5	1606	3131
20	Fish pots	FishP	1.5	2053	3856
21	Squid pots	SquidP	1.5	752	1599
22	Simple hook and lines, handlines	SHL	1	6228	10258
23	Multiple hook and lines	MHL	1	724	3372
24	Troll lines	Troll	1	650	1716
25	Bottom-set longlines	BSLL	1	1373	2344
26	Spear fishing or diving with compressor	Compr	5	100	200
27	Spear fishing or diving	Spear	1	1888	2073
28	Squid jigs and trolls	SqJ	1	3656	8745
29	Scissor nets, push nets	PushN	1	928	1456
30	Mechanized push nets	MPushN	2	80	18
31	Round-haul seine	RHS	2	140	839
32	Scoopnets, cast nets	ScoopN	1	262	261
33	Gleaning, lantern fishing	Glean	1	3939	9814
34	Blast fishing	BlastF	2.5	256	134
	Total			35780	85060

Table S2. Description and sources of inputs for the Visayan Sea baseline Ecopath model parameters

	Group name	Biomass in habitat area	P/B	Q/B	Diets	Catch
1	Seabirds	Estimated by Ecopath	other model (Ainsworth et al., 2006)	other model (Ainsworth et al., 2006)	other model (Ainsworth et al., 2006)	--
2	Sharks (juvenile/young adult)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=0.61$, after Pauly (1980)	original input: 7.13/yr, the weighted mean of empirical estimates of Q/B of representative species after Pauly et al. (1990)	Small sharks (Pitcher et al., 2002)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
3	Rays	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=0.71$, after Pauly (1980)	original input: 7.77/yr, the weighted mean of empirical estimates of Q/B of representative species after Pauly et al. (1990)	Dasyatidae (Kulbicki et al., 2005)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
4	Pelagic predator fish (i)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=1.57$, after Pauly (1980)	original estimate: 21.21/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Auxis</i> spp., <i>Scomberomorus commerson</i> (Bachok et al., 2004)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
5	Pelagic predator fish (a)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=1.12$, after Pauly (1980)	original input: 14.28/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Scomberomorus commerson</i> (Bachok et al., 2004)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
6	Anchovies, sprats	Estimated by Ecopath	initial input based on empirical estimate of $M=2.61$, after Pauly (1980)	original input: 22.40/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Engraulidae (Hajisamae et al., 2004, Milton et al., 1990)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

	Group name	Biomass in habitat area	P/B	Q/B	Diets	Catch
7	Sardines (i)	Estimated by Ecopath	<i>Sardinella</i> spp. in the Visayan Sea (Armada, 2004, Lanzuela et al., 2018)	original estimate: 21.57/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	juvenile <i>Sardinella</i> spp. in the Visayan Sea (Linares, 2020)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
8	Sardines (a)	based on the relationship $F=Y/B^i$	<i>Sardinella</i> spp. in the Visayan Sea (Armada, 2004, Lanzuela et al., 2018)	original input: 19.07/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Sardinella</i> spp. in the Visayan Sea (Mequila and Campos, 2007, Linares, 2020)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
9	Mackerels (i)	Estimated by Ecopath	<i>Rastrelliger</i> spp. in the Visayan Sea (Guanco et al., 2009, Guancho, 1991, Mesa, 2019)	original estimate: 23.76/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	food items for juvenile <i>Rastrelliger kanagurta</i> reported in Fishbase (Froese and Pauly, 2020)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
10	Mackerels (a)	based on the relationship $F=Y/B^i$	<i>Rastrelliger</i> spp. in the Visayan Sea (Guanco et al., 2009, Guancho, 1991, Mesa, 2019)	original input: 19.54/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Rastrelliger</i> sp. in the Visayan Sea (Mequila and Campos, 2007)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
11	Scads (i)	Estimated by Ecopath	<i>Decapterus</i> spp. in the Visayan Sea (Guanco et al., 2009, Mesa, 2019)	original estimate: 21.16/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	juvenile <i>Selaroides leptolepis</i> (Yamashita et al., 1987), juvenile <i>Decapterus</i> (Russell, 1983)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
12	Scads (a)	based on the relationship $F=Y/B^i$	<i>Decapterus</i> spp. in the Visayan Sea (Guanco et al., 2009, Mesa, 2019)	original input: 20.27/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Selaroides leptolepis</i> in the Visayan Sea (Mequila and Campos, 2007); <i>Decapterus</i> spp. (Tiews et al., 1971)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

	Group name	Biomass in habitat area	P/B	Q/B	Diets	Catch
13	Soft-bottom demersal fish (> 30 cm)	Visayan Sea trawl surveys (Aprieto and Villosa, 1979, Armada, 2004, Ampoyos-Arinque, 2018)	Synodontidae in the Visayan Sea (Mesa, 2019)	original input: 10.36/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Synodontidae in the Visayan Sea (Mequila and Campos, 2007)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
14	Soft-bottom demersal fish (< 30 cm)	Visayan Sea trawl surveys (Aprieto and Villosa, 1979, Armada, 2004, Ampoyos-Arinque, 2018)	Mullidae and Nemipteridae in the Visayan Sea (Mesa, 2019)	original input: 16.62/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Leieognathidae, Mullidae and Nemipteridae in the Visayan Sea (Mequila and Campos, 2007)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
15	Reef-associated and inshore predator fish (1)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=1.16$, after Pauly (1980)	original input: 11.32/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Carangidae, Rachycentridae, Sphyraenidae (Bachok et al., 2004, Blaber et al., 1990, Kulbicki et al., 2005, Salini et al., 1994)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
16	Reef-associated and inshore predator fish (2)	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of $M=1.08$, after Pauly (1980)	original input: 11.06/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Haemulidae, Lethrinidae, Lutjanidae, Serranidae (Blaber et al., 1990, Kulbicki et al., 2005, Bachok et al., 2004, Salini et al., 1994)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
17	Reef-associated zoobenthos feeding fish	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of $M=1.98$, after Pauly (1980)	original input: 13.89/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Apogonidae, Labridae, Nemipteridae, Pomacentridae (Sano et al., 1984, Pamintuan and Aliño, 1994, Salini et al., 1994)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
18	Reef-associated grazers/herbivorous fish (1)	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d,	initial input based on empirical estimate of $M=1.46$, after Pauly (1980)	original input: 34.00/yr, the weighted mean of empirical estimates of Q/B of representative species	Scaridae (Ochavillo et al., 1992)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

Group name		Biomass in habitat area	P/B	Q/B	Diets	Catch
19	Reef-associated grazers/herbivorous fish (2)	Campos et al., 2020b) fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of M=2.32, after Pauly (1980)	after Palomares and Pauly (1998) original input: 43.71/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Siganidae (Adhiambo, 2009, Sano et al., 1984)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
20	Reef-associated planktivorous fish	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of M=1.96, after Pauly (1980)	original input: 18.04/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Caesionidae, Pomacentridae (Sano et al., 1984)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
21	Jellyfish	Estimated by Ecopath	other models (Opitz, 1996, Ainsworth et al., 2006)	other models (Opitz, 1996, Ainsworth et al., 2006)	other models (Opitz, 1996, Ainsworth et al., 2006)	--
22	Octopus	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	Octopus (Pauly et al., 1993a)	<i>Octopus</i> spp. (Ambrose, 1984, McQuaid, 1994, Opitz, 1996)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
23	Squids	Estimated by Ecopath	<i>Uroteuthis</i> spp. in the Visayan Sea (Tajolosa, 2011)	Squids (Pauly et al., 1993a)	<i>Uroteuthis</i> spp. (Islam et al., 2018, Tehseen et al., 2019)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
24	Marine crabs	Estimated by Ecopath	<i>Portunus pelagicus</i> in the Visayan Sea (Ingles, 1996, Mesa et al., 2017, Romero, 2009)	Portunidae (Pauly et al., 1993a)	<i>Portunus</i> spp. (de Lestang et al., 2000, Kunsook et al., 2014, Williams, 1982, Ikhwanuddin et al., 2014)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
25	Shrimps and prawns	Estimated by Ecopath	other models (Armada and Bacalso, 2004, Armada et al., 2003)	Penaeidae (Pauly et al., 1993a)	Penaeidae (Tiews et al., 1968, Marte, 1980)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
26	Molluscs	Estimated by Ecopath	exploited scallops in the Visayan Sea (del Norte-Campos and Villarta, 2008, Cabiles et al., 2019)	other models (Alino et al., 1993, Opitz, 1996)	Gastropods and bivalves (Lin et al., 2006, Opitz, 1996, Gabral-Llana, 1986)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

	Group name	Biomass in habitat area	P/B	Q/B	Diets	Catch
27	Other epibenthos	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	Echinoderms, (Pauly et al., 1993a)	various references (Alino et al., 1993, Opitz, 1996)	--
28	Worms, infauna	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	other models (Alino et al., 1993, Opitz, 1996)	Polychaetes (Fauchald and Jumars, 1979)	
29	Sessile benthos	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	other models (Alino et al., 1993, Opitz, 1996)	various references (Pitcher et al., 2002, Alino et al., 1993)	--
30	Zooplankton	Estimated by Ecopath	other models (Bundy and Pauly, 2001, Alino et al., 1993)	other models (Bundy and Pauly, 2001, Alino et al., 1993)	various references (Alino et al., 1993, Guarin, 1991, Campos, 2003)	--
31	Phytoplankton	Chlorophyll-a concentrations extracted for the Visayan Sea area ⁱⁱ	other models (Bundy and Pauly, 2001, Alino et al., 1993)	other models (Bundy and Pauly, 2001, Alino et al., 1993)	--	--
32	Benthic autotrophs	Estimated by Ecopath	other models (Bundy and Pauly, 2001, Alino et al., 1993)	other models (Bundy and Pauly, 2001, Alino et al., 1993)	--	--
33	Detritus	Empirical estimation after Pauly et al. (1993b)	--	--	--	--

ⁱ where Y is the estimated annual catch (t/km²/year) and F is the fishing mortality rate (/year)

ⁱⁱ https://coastwatch.pfeg.noaa.gov/erddap/griddap/NCEP_Global_Best.html (accessed 2020); chlorophyll-a to phytoplankton biomass conversion factors from Opitz (1996)

Table S3. ECOIND trait inputs.

Groups and taxa	Organism	Ecology	Catch contribution	IUCN conservation status ¹	VI (0, 100) ²	Mean length (cm)	Maximum length (cm)
Seabirds							
Seagulls, terns	Birds	Land-based	*	*	*	*	*
Sharks, juveniles-young adult							
Carcharhinidae	Fishes	Benthopelagic	0.89	*	*	56.0	75.0
Hemiscilliidae	Fishes	Reef-associated	0.05	*	*	45.0	*
Scyliorhinidae	Fishes	Reef-associated	0.05	*	*	45.0	*
Stegostomidae	Fishes	Reef-associated	0.01	*	*	45.0	*
Rays							
<i>Dasyatis kuhlii</i>	Fishes	Reef-associated	0.62	Data deficient	52	30 (DW)	35 (DW)
Other Dasyatidae	Fishes	Reef-associated	0.25	*	*	26 (DW)	32 (DW)
<i>Rhinoptera javanica</i>	Fishes	Reef-associated	0.12	Endangered	86	31 (DW)	45 (DW)
<i>Aetobatus narinari</i>	Fishes	Benthopelagic	0.01	Endangered	75	49 (DW)	100 (DW)
Pelagic, intermediate							
<i>Scomberomorus commerson</i>	Fishes	Pelagic-neritic	1.00	Near threatened	52	50.0	124.8
Anchovies, sprats, silversides							
<i>Stolephorus commersonnii</i>	Fishes	Pelagic-neritic	0.34	Least concern	14	9.2	12.4
<i>Stolephorus</i> spp.	Fishes	Pelagic-neritic	0.23	*	*	12.7	15.0
<i>Stolephorus indicus</i>	Fishes	Pelagic-neritic	0.22	Least concern	13	7.7	11.2
<i>Spratelloides gracilis</i>	Fishes	Pelagic-neritic	0.16	Least concern	10	7.5	9.5
Atherinidae	Fishes	Reef-associated	0.01	*	*	10.5	14.0
<i>Encrasicholina heteroloba</i>	Fishes	Reef-associated	0.00	Least concern	10	5.2	8.0
Sardines							
<i>Sardinella gibbosa</i>	Fishes	Pelagic-neritic	0.48	Least concern	15	15.0	22.0
<i>Sardinella fimbriata</i>	Fishes	Pelagic-neritic	0.40	Least concern	15	13.7	19.3
<i>Dussumieria acuta</i>	Fishes	Pelagic-neritic	0.05	Least concern	20	13.2	14.3
<i>Amblygaster clupeioides</i>	Fishes	Reef-associated	0.03	Least concern	18	12.5	22.0
<i>Sardinella lemuru</i>	Fishes	Pelagic-neritic	0.02	Near threatened	17	14.0	24.3
<i>Herklosichthys</i> sp.	Fishes	Pelagic-neritic	0.01	*	*	7.4	8.5
<i>Sardinella albella</i>	Fishes	Reef-associated	0.01	Least concern	10	9.5	15.0
Mackerels							
<i>Rastrelliger kanagurta</i>	Fishes	Pelagic-neritic	0.58	Data deficient	15	25.7	34.0
<i>Rastrelliger brachysoma</i>	Fishes	Pelagic-neritic	0.41	Data deficient	18	21.0	27.2
<i>Rastrelliger faughni</i>	Fishes	Pelagic-neritic	0.01	Data deficient	11	16.7	17.5
Scads							
<i>Selaroides leptolepis</i>	Fishes	Demersal	0.50	Least concern	14	13.2	18.1
<i>Decapterus muruadsi</i>	Fishes	Pelagic-oceanic	0.23	Least concern	42	19.0	22.0

¹ <https://www.iucnredlist.org/>

² VI - Vulnerability Index sourced from Fishbase (<https://fishbase.mnhn.fr/search.php>) and Sealifebase (<https://www.sealifebase.se/search.php>)

Groups and taxa	Organism	Ecology	Catch contribution	IUCN conservation status ¹	VI (0, 100) ²	Mean length (cm)	Maximum length (cm)
<i>Atule mate</i>	Fishes	Pelagic-neritic	0.16	Least concern	19	12.5	21.5
<i>Megalaspis cordyla</i>	Fishes	Pelagic-neritic	0.03	Least concern	29	28.2	31.0
<i>Selar crumenophthalmus</i>	Fishes	Pelagic-neritic	0.03	Least concern	39	17.7	22.8
<i>Alepes melanoptera</i>	Fishes	Pelagic-neritic	0.02	Least concern	17	15.8	25.0
<i>Selar boops</i>	Fishes	Reef-associated	0.02	Least concern	18	11.7	16.0
<i>Alepes djedaba</i>	Fishes	Reef-associated	0.01	Least concern	25	14.5	25.0
<i>Decapterus macrosoma</i>	Fishes	Pelagic-neritic	0.01	Least concern	17	21.0	31.0
Demersal fish, piscivorous							
<i>Platycephalus indicus</i>	Fishes	Demersal	0.40	Data deficient	35	27.8	48.0
<i>Saurida tumbil</i>	Fishes	Demersal	0.20	Least concern	38	28.7	35.0
<i>Eleutheronema tetradactylum</i>	Fishes	Demersal	0.13	Not evaluated	90	32.0	50.0
<i>Chirocentrus dorab</i>	Fishes	Demersal	0.05	Least concern	73	35.5	40.6
<i>Trichiurus lepturus</i>	Fishes	Benthopelagic	0.04	Least concern	45	72.7	89.3
<i>Pseudorhombus dupliciocellatus</i>	Fishes	Demersal	0.03	Least concern	39	26.4	32.0
<i>Saurida undosquamis</i>	Fishes	Demersal	0.03	Least concern	28	30.6	35.0
<i>Polynemus</i> sp.	Fishes	Demersal	0.03	Not evaluated	90	37.1	52.2
<i>Pardachirus pavoninus</i>	Fishes	Demersal	0.03	Least concern	15	26.7	31.5
<i>Megalops cyprinoides</i>	Fishes	Benthopelagic	0.02	Data deficient	90	30.0	45.0
<i>Lates calcarifer</i>	Fishes	Demersal	0.01	Least concern	70	33.2	51.4
<i>Elates ransonnetti</i>	Fishes	Demersal	0.01	Not evaluated	10	27.5	41.6
<i>Acanthopagrus berda</i>	Fishes	Demersal	0.01	Least concern	42	25.0	40.0
<i>Clarias</i> sp.	Fishes	Demersal	0.01	*	*	*	*
<i>Cynoglossus puncticeps</i>	Fishes	Demersal	0.01	Least concern	24	28.0	32.6
<i>Inegocia japonica</i>	Fishes	Demersal	0.00	Least concern	25	37.0	42.0
<i>Umbrina</i> sp.	Fishes	Demersal	0.00	*	*	*	*
<i>Elops hawaiiensis</i>	Fishes	Benthopelagic	0.00	Data deficient	88	25.0	70.0
Demersal fish, small							
<i>Leiognathus splendens</i>	Fishes	Demersal	0.18	Least concern	13	9.6	15.0
<i>Scolopsis taenioptera</i>	Fishes	Demersal	0.14	Least concern	27	15.0	23.0
<i>Upeneus japonicus</i>	Fishes	Demersal	0.09	Not evaluated	22	14.2	25.9
<i>Nemipterus hexodon</i>	Fishes	Demersal	0.06	Least concern	27	16.9	24.8
<i>Ellochelon vaigiensis</i>	Fishes	Demersal	0.06	Least concern	45	16.0	38.0
<i>Pelates quadrilineatus</i>	Fishes	Demersal	0.05	Not evaluated	20	14.5	17.4
<i>Gerres erythrourus</i>	Fishes	Demersal	0.05	Least concern	20	16.3	23.6
<i>Sillago sihama</i>	Fishes	Demersal	0.05	Least concern	24	18.0	25.9
<i>Leiognathus elongatus</i>	Fishes	Demersal	0.04	Not evaluated	10	7.3	8.3
<i>Gerres</i> spp.	Fishes	Demersal	0.04	*	*	13.4	18.9
<i>Leiognathus elongatus</i>	Fishes	Demersal	0.04	Not evaluated	10	7.3	8.3
<i>Leiognathus lineolatus</i>	Fishes	Demersal	0.03	Not evaluated	13	11.4	15.1
<i>Eubleekeria rapsoni</i>	Fishes	Demersal	0.03	Not evaluated	10	7.9	10.1
<i>Gerres filamentosus</i>	Fishes	Demersal	0.02	Least concern	32	19.8	26.3

Groups and taxa	Organism	Ecology	Catch contribution	IUCN conservation status ¹	VI (0, 100) ²	Mean length (cm)	Maximum length (cm)
<i>Leiognathus bindus</i>	Fishes	Demersal	0.02	Not evaluated	24	9.0	13.2
<i>Priacanthus tayenus</i>	Fishes	Benthic	0.02	Least concern	23	16.9	21.4
<i>Oxyurichthys microlepis</i>	Fishes	Demersal	0.01	Least concern	10	14.2	16.3
Cepolidae	Fishes	Demersal	0.01	*	*	*	26.3
<i>Gerres oyena</i>	Fishes	Demersal	0.01	Least concern	20	12.3	16.3
<i>Secutor ruconius</i>	Fishes	Demersal	0.01	Not evaluated	12	3.5	8.6
<i>Parascolopsis eriomma</i>	Fishes	Demersal	0.01	Least concern	25	19.1	22.9
<i>Upeneus</i> spp.	Fishes	Demersal	0.01	*	*	17.6	21.9
<i>Upeneus tragula</i>	Fishes	Demersal	0.01	Least concern	17	14.1	19.1
<i>Leiognathus equulus</i>	Fishes	Demersal	0.00	Least concern	17	6.4	18.4
<i>Therapon theraps</i>	Fishes	Demersal	0.00	Least concern	24	17.6	20.4
<i>Nemipterus bathybius</i>	Fishes	Demersal	0.00	Least concern	30	14.8	25.0
<i>Therapon puta</i>	Fishes	Benthopelagic	0.00	Not evaluated	20	13.3	16.4
<i>Leiognathus leuciscus</i>	Fishes	Demersal	0.00	Least concern	13	8.9	17.8
<i>Upeneus vittatus</i>	Fishes	Demersal	0.00	Least concern	25	19.3	25.4
<i>Therapon jarbua</i>	Fishes	Demersal	0.00	Least concern	37	13.7	18.6
<i>Gazza minuta</i>	Fishes	Demersal	0.00	Least concern	21	7.0	14.6
<i>Nemipterus marginatus</i>	Fishes	Demersal	0.00	Least concern	22	12.0	15.3
<i>Parascolopsis</i> spp.	Fishes	Demersal	0.00	*	*	15.3	20.5
<i>Upeneus sulphureus</i>	Fishes	Demersal	0.00	Least concern	15	12.7	15.2
<i>Upeneus moluccensis</i>	Fishes	Demersal	0.00	Least concern	22	13.3	15.3
<i>Nemipterus peronii</i>	Fishes	Demersal	0.00	Least concern	26	20.7	23.9
<i>Nemipterus japonicus</i>	Fishes	Demersal	0.00	Least concern	21	24.1	29.1
<i>Priacanthus macracanthus</i>	Fishes	Benthic	0.00	Least concern	18	19.0	22.0
<i>Nemipterus nematophorus</i>	Fishes	Demersal	0.00	Least concern	24	19.8	24.6
<i>Aurigequula fasciata</i>	Fishes	Demersal	0.00	Least concern	11	12.0	15.0
<i>Karalla daura</i>	Fishes	Demersal	0.00	Not evaluated	11	7.4	12.2
<i>Nemipterus virgatus</i>	Fishes	Demersal	0.00	Vulnerable	35	21.9	23.9
<i>Upeneus sundaicus</i>	Fishes	Demersal	0.00	Least concern	12	13.9	18.9
Jacks, barracudas, needlefish							
<i>Sphyrna jello</i>	Fishes	Reef-associated	0.39	Not evaluated	75	34.1	*
<i>Tylosurus acus melanotus</i>	Fishes	Reef-associated	0.13	Not evaluated	60	56.2	65.0
<i>Sphyrna barracuda</i>	Fishes	Reef-associated	0.10	Least concern	79	48.0	85.0
<i>Strongylura leiura</i>	Fishes	Reef-associated	0.07	Not evaluated	60	53.0	97.0
<i>Sphyrna forsteri</i>	Fishes	Reef-associated	0.06	Not evaluated	61	30.0	50.0
<i>Tylosurus crocodilus</i>	Fishes	Reef-associated	0.04	Least concern	47	60.0	88.6
<i>Caranx</i> spp.	Fishes	Reef-associated	0.03	*	*	25.3	*
<i>Formio niger</i>	Fishes	Reef-associated	0.03	Least concern	30	24.6	40.0
<i>Alectis indicus</i>	Fishes	Reef-associated	0.03	Least concern	62	25.0	47.0
<i>Carangoides malabaricus</i>	Fishes	Reef-associated	0.02	Least concern	25	32.6	35.0
<i>Caranx sexfasciatus</i>	Fishes	Reef-associated	0.02	Least concern	45	35.5	*
<i>Uraspis helvola</i>	Fishes	Reef-associated	0.02	Least concern	43	35.0	43.0

Groups and taxa	Organism	Ecology	Catch contribution	IUCN conservation status ¹	VI (0, 100) ²	Mean length (cm)	Maximum length (cm)
<i>Sphyraena obtusata</i>	Fishes	Reef-associated	0.01	Not evaluated	25	28.0	35.0
<i>Ablennes hians</i>	Fishes	Reef-associated	0.01	Least concern	40	57.3	60.0
<i>Scomberoides tol</i>	Fishes	Reef-associated	0.01	Least concern	24	41.0	60.0
<i>Caranx ignobilis</i>	Fishes	Reef-associated	0.01	Least concern	74	34.3	*
<i>Gnathanodon speciosus</i>	Fishes	Reef-associated	0.01	Least concern	43	36.0	66.0
<i>Carangoides fulvoguttatus</i>	Fishes	Reef-associated	0.00	Least concern	80	57.7	*
<i>Strongylura incisa</i>	Fishes	Reef-associated	0.00	Not evaluated	67	60.0	115.0
<i>Trachinotus bailloni</i>	Fishes	Reef-associated	0.00	Least concern	44	35.0	60.0
<i>Rachycentron canadum</i>	Fishes	Reef-associated	0.00	Least concern	42	72.0	107.0
<i>Scomberoides commersonianus</i>	Fishes	Reef-associated	0.00	Least concern	48	60.0	110.0
<i>Elagatis bipinnulatus</i>	Fishes	Reef-associated	0.00	Least concern	51	40.0	90.0
<i>Ulua mentalis</i>	Fishes	Reef-associated	0.00	Least concern	60	35.0	60.0
<i>Carangoides dinema</i>	Fishes	Reef-associated	0.00	Least concern	32	27.8	40.0
<i>Sphyraena flavicauda</i>	Fishes	Reef-associated	0.00	Not evaluated	33	28.0	40.0
Reef-associated carnivorous/piscivorous fish							
<i>Lethrinus</i> spp.	Fishes	Reef-associated	0.22	*	*	22.3	35.5
<i>Lethrinus lentjan</i>	Fishes	Reef-associated	0.16	Least concern	25	23.3	31.1
<i>Plectorhynchus pictus</i>	Fishes	Reef-associated	0.13	Least concern	52	41.9	52.8
<i>Epinephelus</i> spp.	Fishes	Reef-associated	0.12	*	*	24.0	36.1
<i>Lethrinus reticulatus</i>	Fishes	Reef-associated	0.04	Least concern	30	16.0	20.0
<i>Epinephelus merra</i>	Fishes	Reef-associated	0.04	Least concern	29	26.0	34.0
<i>Conger</i> spp.	Fishes	Reef-associated	0.03	*	*	44.0	64.0
<i>Lethrinus erythropterus</i>	Fishes	Reef-associated	0.03	Least concern	40	15.0	30.0
<i>Lutjanus</i> spp.	Fishes	Reef-associated	0.02	*	*	23.9	45.8
<i>Cephalopholis boenak</i>	Fishes	Reef-associated	0.02	Least concern	31	22.0	30.0
<i>Epinephelus sexfasciatus</i>	Fishes	Reef-associated	0.02	Least concern	26	25.0	40.0
<i>Lethrinus miniatus</i>	Fishes	Reef-associated	0.02	Least concern	41	28.5	36.4
<i>Plectorhynchus</i> spp.	Fishes	Reef-associated	0.02	*	*	30.7	35.7
<i>Plectropomus</i> spp.	Fishes	Reef-associated	0.02	*	*	21.6	37.1
<i>Cephalopholis argus</i>	Fishes	Reef-associated	0.02	Least concern	44	35.0	40.0
<i>Lutjanus monostigma</i>	Fishes	Reef-associated	0.01	Least concern	40	26.0	31.6
<i>Cephalopholis sexmaculatus</i>	Fishes	Reef-associated	0.01	Least concern	33	26.0	40.0
<i>Epinephelus corallicola</i>	Fishes	Reef-associated	0.01	Least concern	39	30.0	40.0
<i>Lethrinus erythracanthus</i>	Fishes	Reef-associated	0.01	Least concern	46	29.3	50.0
<i>Lethrinus harak</i>	Fishes	Reef-associated	0.01	Least concern	31	19.0	30.0
<i>Lethrinus obsoletus</i>	Fishes	Reef-associated	0.01	Least concern	33	29.3	58.1
<i>Plectorhynchus chaetodonoides</i>	Fishes	Reef-associated	0.01	Not evaluated	49	29.3	32.2
<i>Gymnothorax undulatus</i>	Fishes	Reef-associated	0.00	Least concern	90	*	94.0
<i>Lethrinus nebulosus</i>	Fishes	Reef-associated	0.00	Least concern	47	28.9	37.2
<i>Lethrinus genivittatus</i>	Fishes	Reef-associated	0.00	Least concern	17	20.3	25.3
<i>Epinephelus quoyanus</i>	Fishes	Reef-associated	0.00	Least concern	30	32.7	40.0

Groups and taxa	Organism	Ecology	Catch contribution	IUCN conservation status ¹	VI (0, 100) ²	Mean length (cm)	Maximum length (cm)
<i>Gymnothorax punctatofasciatus</i>	Fishes	Reef-associated	0.00	Not evaluated	40	*	95.0
<i>Lutjanus vitta</i>	Fishes	Reef-associated	0.00	Least concern	23	18.8	28.0
<i>Bathycongrus retrotinctus</i>	Fishes	Reef-associated	0.00	Data deficient	42	51.0	54.0
<i>Congersox talabon</i>	Fishes	Reef-associated	0.00	Least concern	52	50.0	88.9
<i>Echidna nebulosa</i>	Fishes	Reef-associated	0.00	Least concern	60	50.0	82.2
<i>Epinephelus bleekeri</i>	Fishes	Reef-associated	0.00	Data deficient	60	25.2	40.0
<i>Lutjanus carponotatus</i>	Fishes	Reef-associated	0.00	Least concern	26	22.3	40.0
<i>Lutjanus decussatus</i>	Fishes	Reef-associated	0.00	Least concern	25	14.7	30.0
Reef-associated zoobenthos-feed. fish							
<i>Pentapodus setosus</i>	Fishes	Reef-associated	0.20	Not evaluated	12	15.2	20.3
<i>Apogonidae</i>	Fishes	Reef-associated	0.19	*	*	9.7	13.4
<i>Plotosus lineatus</i>	Fishes	Reef-associated	0.19	Not evaluated	28	18.0	25.4
<i>Pentapodus emeryii</i>	Fishes	Reef-associated	0.13	Least concern	25	15.9	17.0
<i>Monacanthus spp.</i>	Fishes	Reef-associated	0.07	*	*	11.8	17.6
<i>Parupeneus spp.</i>	Fishes	Reef-associated	0.03	*	*	16.8	28.5
<i>Pentapodus spp.</i>	Fishes	Reef-associated	0.03	*	*	15.2	20.8
<i>Balistoides viridescens</i>	Fishes	Reef-associated	0.02	Least concern	50	15.9	32.7
<i>Parupeneus barberinus</i>	Fishes	Reef-associated	0.02	Least concern	40	17.7	19.2
<i>Parapercis sp.</i>	Fishes	Reef-associated	0.01	*	*	12.0	15.0
<i>Scolopsis spp.</i>	Fishes	Reef-associated	0.01	*	*	13.0	25.0
<i>Parupeneus heptacanthus</i>	Fishes	Reef-associated	0.01	Least concern	38	22.0	28.5
<i>Cheilinus undulatus</i>	Fishes	Reef-associated	0.01	Endangered	74	24.0	*
<i>Scolopsis lineata</i>	Fishes	Reef-associated	0.01	Least concern	15	13.0	20.0
<i>Scolopsis aurata</i>	Fishes	Reef-associated	0.01	Least concern	16	12.0	26.0
<i>Cheilinus fasciatus</i>	Fishes	Reef-associated	0.01	Least concern	39	18.5	30.0
<i>Lagocephalus lunaris</i>	Fishes	Reef-associated	0.01	Least concern	42	13.5	30.0
<i>Pomacanthus spp.</i>	Fishes	Reef-associated	0.01	*	*	7.0	10.0
<i>Chaetodon octofasciatus</i>	Fishes	Reef-associated	0.01	Least concern	10	6.4	8.0
<i>Bodianus bilunulatus</i>	Fishes	Reef-associated	0.00	Least concern	42	19.1	21.4
<i>Bodianus spp.</i>	Fishes	Reef-associated	0.00	*	*	18.3	25.0
<i>Choerodon anchorago</i>	Fishes	Reef-associated	0.00	Least concern	40	21.5	30.0
<i>Cheilinus trilobatus</i>	Fishes	Reef-associated	0.00	Least concern	35	10.7	26.0
<i>Cheilinus spp.</i>	Fishes	Reef-associated	0.00	*	*	17.0	20.3
<i>Mulloidichthys flavolineatus</i>	Fishes	Reef-associated	0.00	Least concern	39	15.7	17.4
<i>Cheilinus bimaculatus</i>	Fishes	Reef-associated	0.00	Least concern	10	13.0	18.7
<i>Gymnocranius griseus</i>	Fishes	Reef-associated	0.00	Least concern	38	13.3	*
<i>Labridae</i>	Fishes	Reef-associated	0.00	*	*	10.6	12.1
<i>Diodon holocanthus</i>	Fishes	Reef-associated	0.00	Least concern	27	6.7	*
<i>Anampses caeruleopunctatus</i>	Fishes	Reef-associated	0.00	Least concern	32	19.8	*
<i>Cheilinus diagrammus</i>	Fishes	Reef-associated	0.00	Least concern	39	22.1	34.7
<i>Choerodon schoenleinii</i>	Fishes	Reef-associated	0.00	Near threatened	39	25.5	*

Groups and taxa	Organism	Ecology	Catch contribution	IUCN conservation status ¹	VI (0, 100) ²	Mean length (cm)	Maximum length (cm)
<i>Chelmon rostratus</i>	Fishes	Reef-associated	0.00	Least concern	10	10.3	15.8
<i>Cheilodipterus quinquelineatus</i>	Fishes	Reef-associated	0.00	Least concern	10	8.5	10.5
<i>Ostorhinchus margaritophorus</i>	Fishes	Reef-associated	0.00	Least concern	10	6.5	*
<i>Cheilio enermis</i>	Fishes	Reef-associated	0.00	Least concern	44	*	28.2
<i>Halichoeres melanurus</i>	Fishes	Reef-associated	0.00	Least concern	10	*	12.9
Reef-associated herbivorous fish							
<i>Leptoscarus vaigiensis</i>	Fishes	Reef-associated	0.28	Least concern	25	22.6	29.50
<i>Scarus ghobban</i>	Fishes	Reef-associated	0.05	Least concern	36	18.2	20.14
<i>Chlorurus bleekeri</i>	Fishes	Reef-associated	0.05	Least concern	39	17.8	25.0
<i>Cetoscarus bicolor</i>	Fishes	Reef-associated	0.02	Least concern	34	22.36	36.71
Scaridae	Fishes	Reef-associated	0.60	*	*	17.15	27.78
Rabbitfish							
<i>Siganus canaliculatus</i>	Fishes	Reef-associated	0.55	Least concern	18	16.58	19.93
<i>Siganus corallinus</i>	Fishes	Reef-associated	0.02	Least concern	33	18.57	19.53
<i>Siganus guttatus</i>	Fishes	Reef-associated	0.01	Least concern	19	19.64	26.04
<i>Siganus javus</i>	Fishes	Reef-associated	0.04	Least concern	41	21.22	25.87
<i>Siganus lineatus</i>	Fishes	Reef-associated	0.00	Least concern	25	16.69	26.69
<i>Siganus spinus</i>	Fishes	Reef-associated	0.01	Least concern	14	9.29	19.29
<i>Siganus</i> spp.	Fishes	Reef-associated	0.36	*	*	16.51	25.87
Reef-associated planktivorous fish							
<i>Caesio cuning</i>	Fishes	Reef-associated	0.51	Least concern	44	19.5	26.0
<i>Caesio</i> spp.	Fishes	Reef-associated	0.28	*	*	17.0	25.0
<i>Pterocaesio tile</i>	Fishes	Reef-associated	0.13	Least concern	20	17.0	19.0
<i>Pterocaesio diagramma</i>	Fishes	Reef-associated	0.07	Least concern	31	15.0	21.6
Jellyfishes							
Jellyfish	Invertebrates	Benthopelagic	1.00	*	*	*	*
Octopus							
<i>Octopus</i> spp.	Invertebrates	Benthic	1.00	*	*	5.6 (ML)	6.17 (ML)
Squids and cuttlefish							
<i>Uroteuthis duvaucellii</i>	Invertebrates	Demersal	0.70	Data deficient	14	15 (ML)	32 (ML)
<i>Sepioteuthis lessoniana</i>	Invertebrates	Demersal	0.16	Data deficient	19	*	40.02 (ML)
<i>Sepia</i> spp.	Invertebrates	Demersal	0.04	*	*	11.84 (ML)	22.9 (ML)
<i>Uroteuthis edulis</i>	Invertebrates	Demersal	0.10	Data deficient	30	13.39 (ML)	33 (ML)
Marine crabs							
<i>Portunus pelagicus</i>	Invertebrates	Demersal	0.85	Not evaluated	10	11.3 (CW)	20 (CW)
<i>Portunus sanguinolentus</i>	Invertebrates	Demersal	0.06	Not evaluated	10	9.6 (CW)	20 (CW)
<i>Charybdis feriata</i>	Invertebrates	Benthic	0.03	Not evaluated	10	7 (CW)	14.5 (CW)
<i>Scylla</i> spp.	Invertebrates	Benthic	0.00	*	*	10.6 (CW)	*
other marine crabs	Invertebrates	Benthic	0.06	*	*	8.7 (CW)	*
Shrimps							
<i>Metapenaeus ensis</i>	Invertebrates	Benthic	0.51	Not evaluated	10	8.4 (TL)	*

Groups and taxa	Organism	Ecology	Catch contribution	IUCN conservation status ¹	VI (0, 100) ²	Mean length (cm)	Maximum length (cm)
<i>Penaeus monodon</i>	Invertebrates	Benthic	0.17	Not evaluated	17	15.2 (TL)	18.5 (TL)
other Penaeidae	Invertebrates	Benthic	0.33	*	*	11.6 (TL)	*
Shelled molluscs							
Bivalves (pen shells, scallops, oysters, clams)	Invertebrates	Benthic	0.50	*	*	*	*
Gastropods (conchs, top shells, turbans, cone shells, abalone)	Invertebrates	Benthic	0.50	*	*	*	*
Other macrobenthos							
Echinoderms	Invertebrates	Benthic	0.53	*	*	*	*
<i>Thenus orientalis</i>	Invertebrates	Benthic	0.13	Least concern	15	8.5 (CL)	*
<i>Panulirus</i> spp.	Invertebrates	Benthic	0.10	*	*	9 (CL)	*
Stomatopoda	Invertebrates	Benthic	0.24	*	*	15 (TL)	*
Worms, infauna							
Polychaetes	Invertebrates	Benthic	*	*	*	*	*
Flatworms	Invertebrates	Benthic	*	*	*	*	*

*no or insufficient data; DW – disc width; ML – mantle length; TL – total length; CL – carapace length; CW – carapace width

SUPPLEMENTARY FIGURES

Figure S1 Trends in fishing effort of the commercial fleet types applied to drive the Visayan Sea model simulations

Commercial (> 3 G.T.)

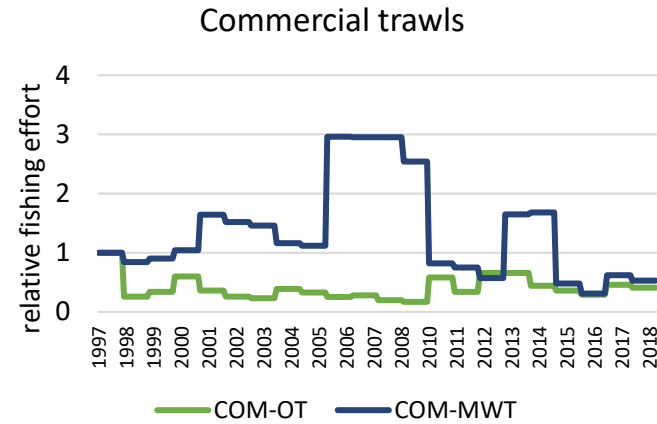
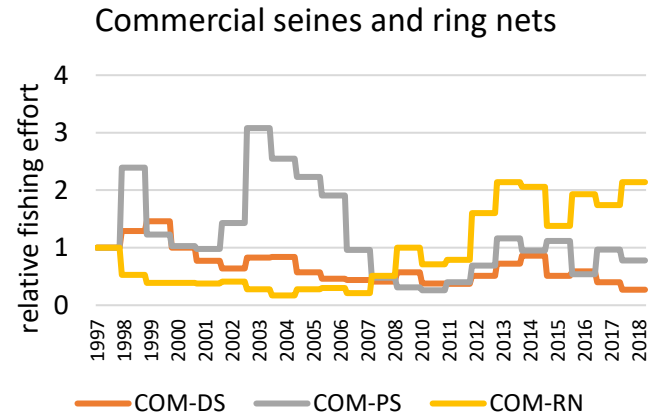
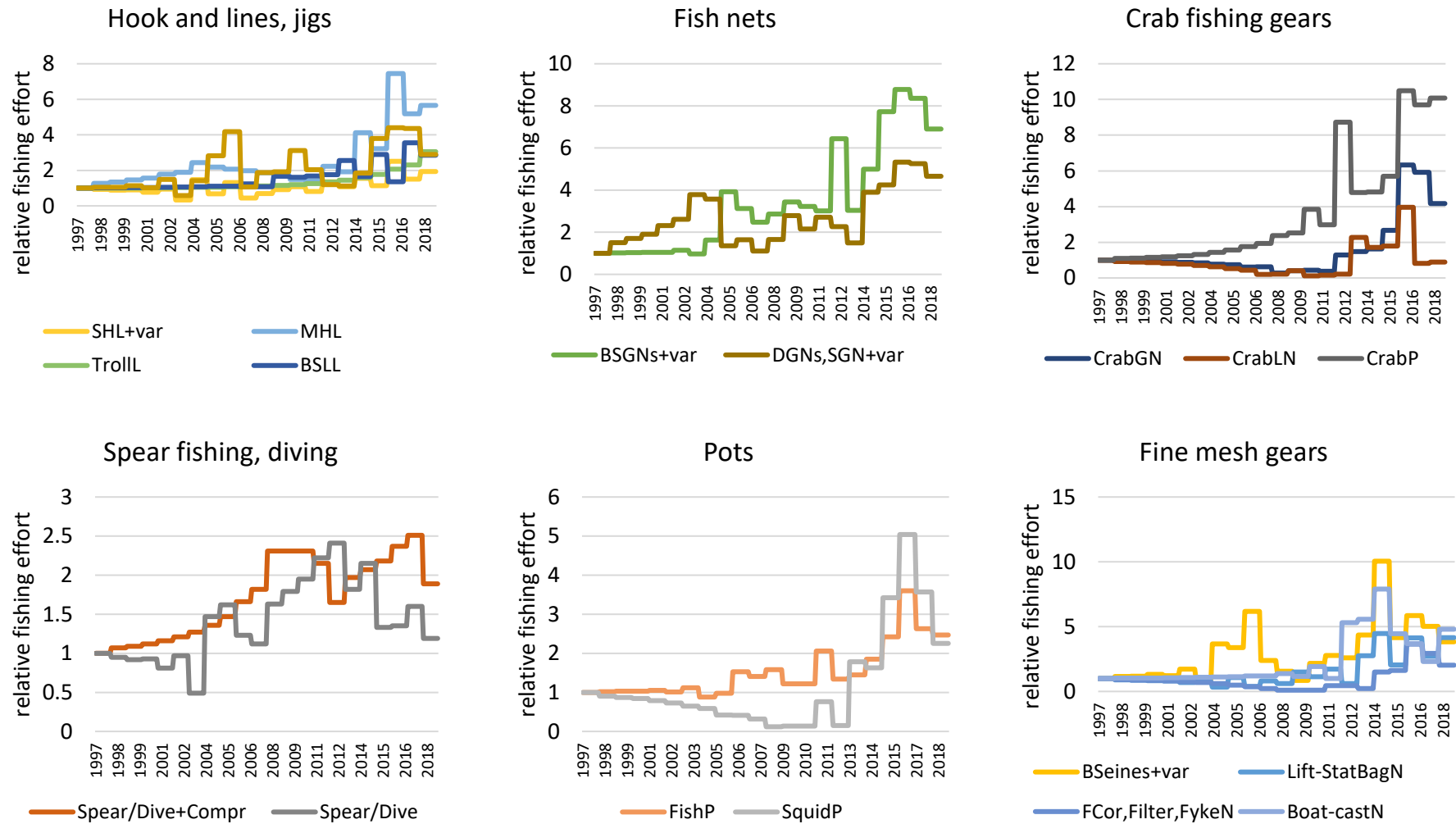


Figure S2. Trends in fishing effort of the municipal fleet types applied to drive the Visayan Sea model simulations

Municipal (< 3 G.T.)



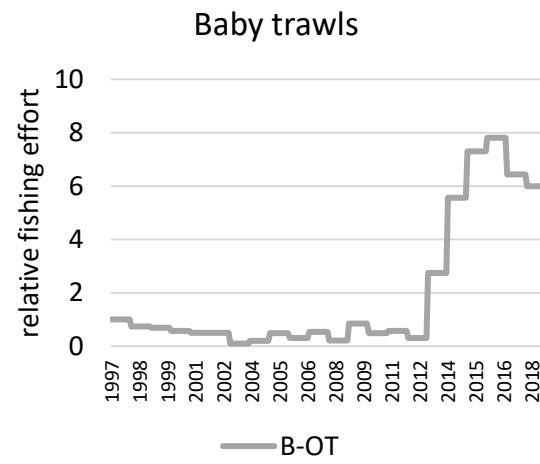
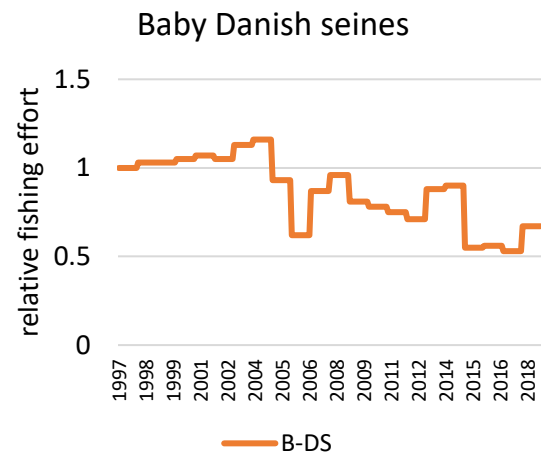
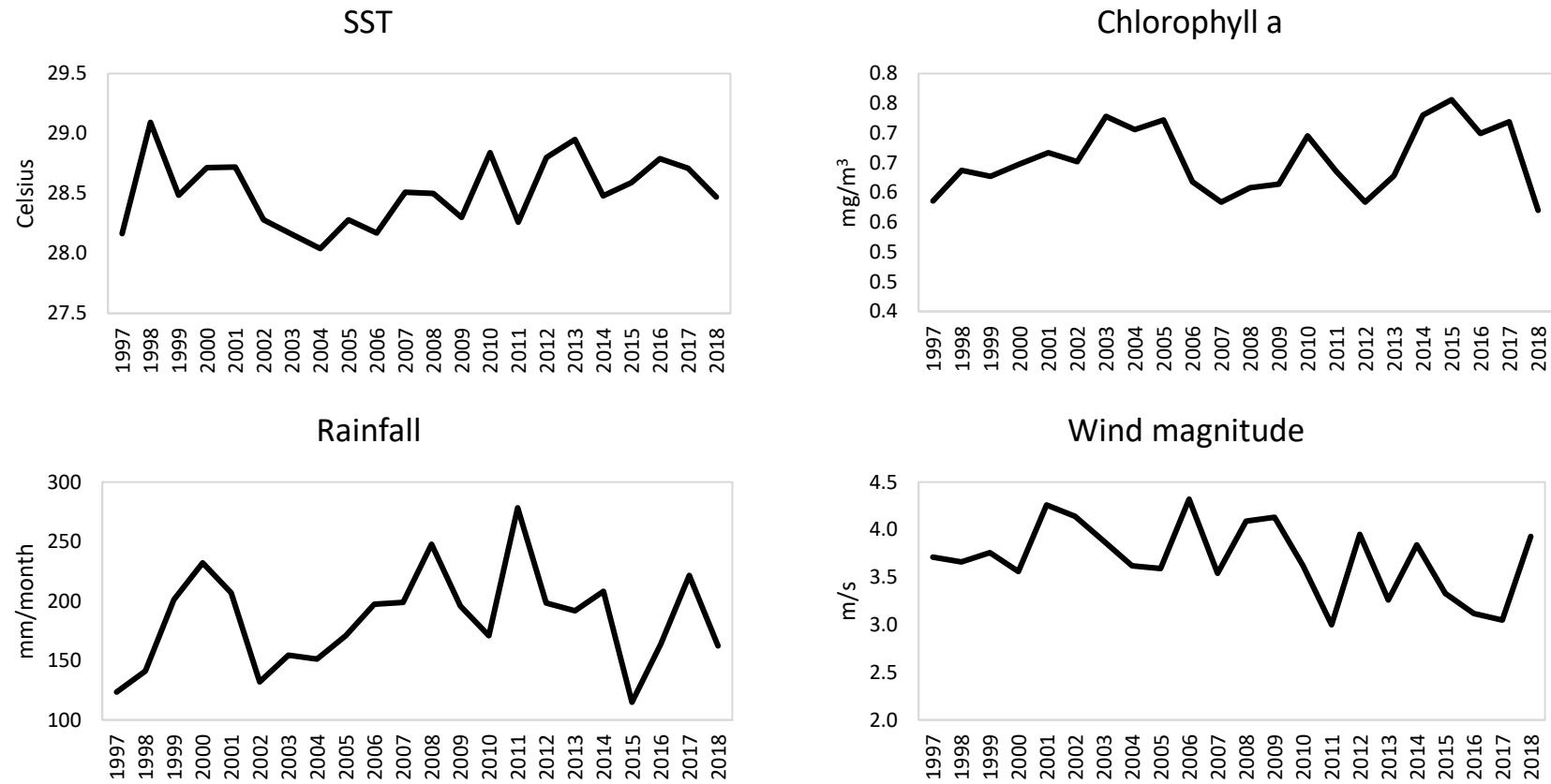
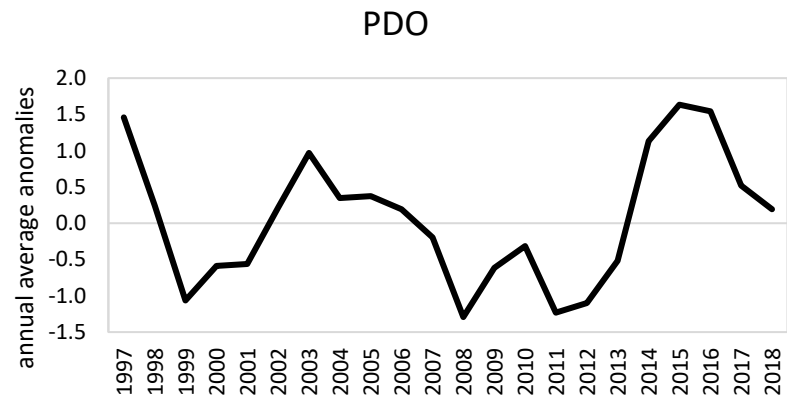
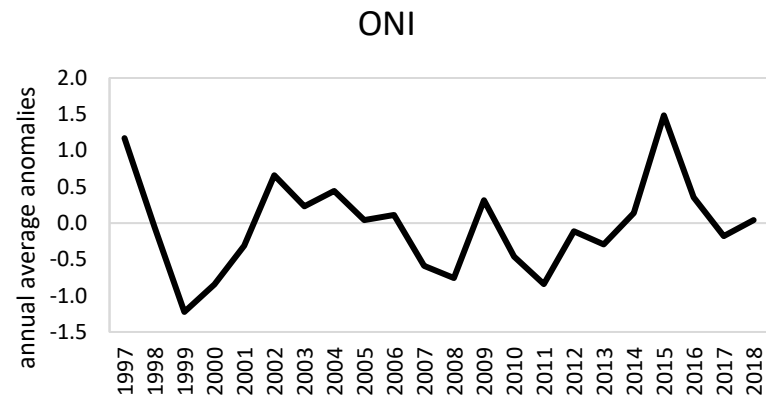
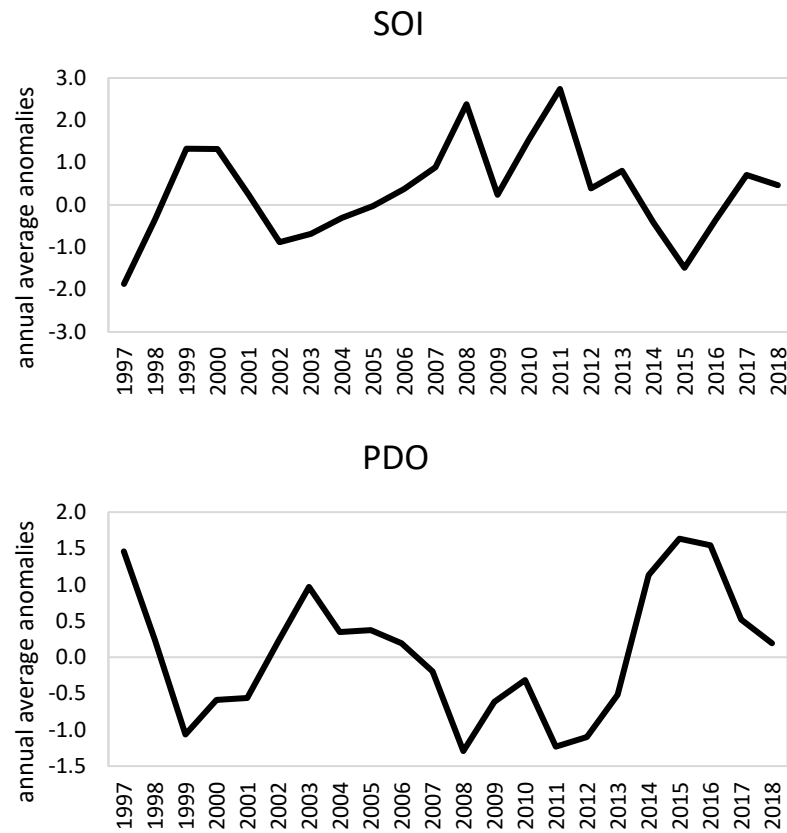


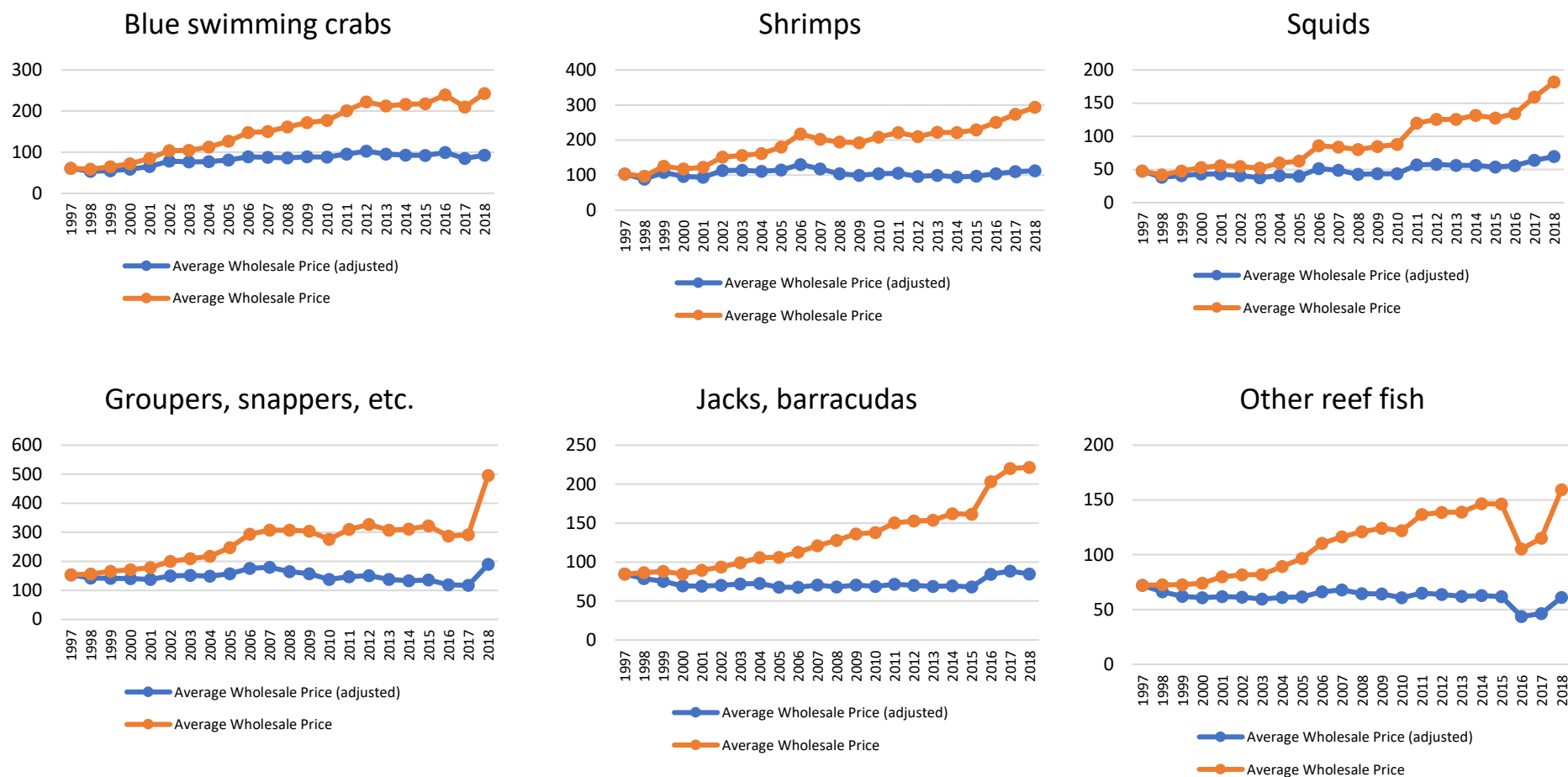
Figure S3. Trends in environmental parameters used as forcing functions in the Visayan Sea model



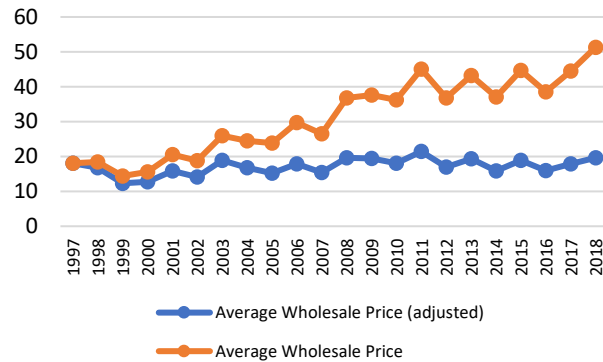


Time series information of remotely-sensed local SST, chlorophyll-a concentration, rainfall, and wind magnitude were sourced from NOAA's ERDDAP (<https://coastwatch.pfeg.noaa.gov/erddap/griddap/>), an online electronic repository and aggregator of spatio-temporal oceanographic data (Simons, 2020). The Southern Oscillation Index (SOI), Oceanic Niño Index (ONI), and Pacific Decadal Oscillation (PDO) were sourced from NOAA's Physical Sciences Laboratory (<https://psl.noaa.gov/>).

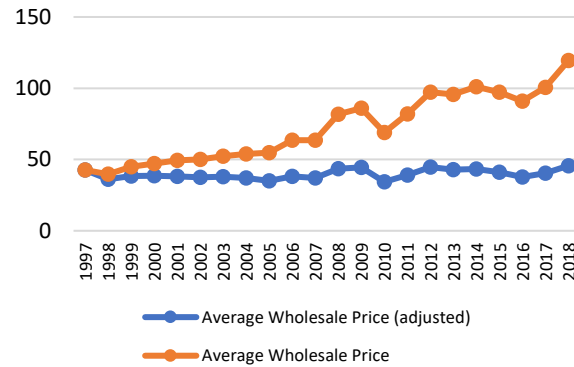
Figure S4. Trends in unadjusted and adjusted wholesale prices (Php/kg) used to estimate the catch values of commercially important fish and invertebrates from the Visayan Sea (fish prices were sourced from the Philippine Statistics Authority for Regions 5,6,7)



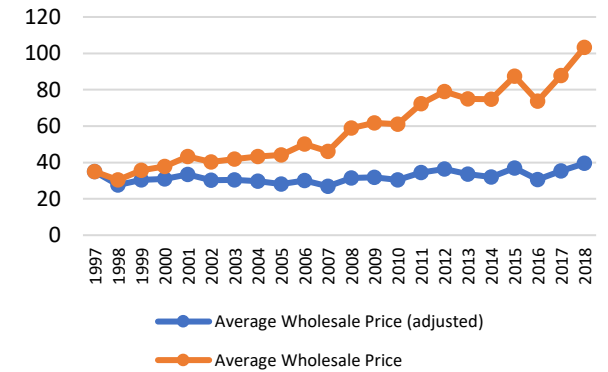
Sardines



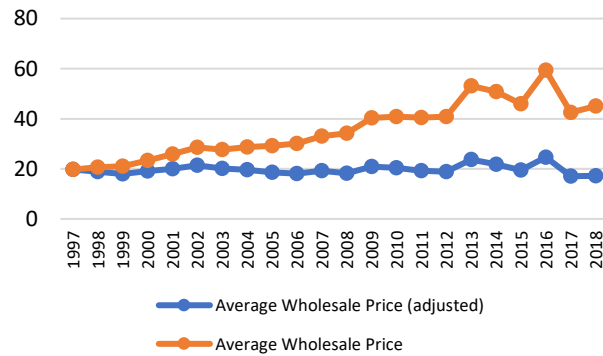
Mackerels



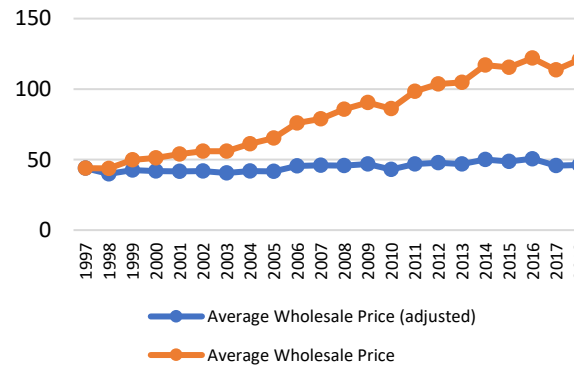
Scads



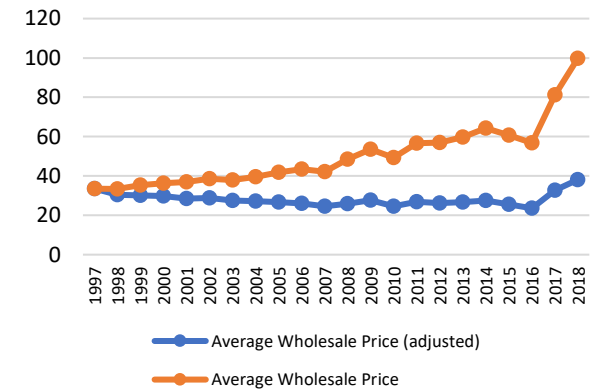
Anchovies



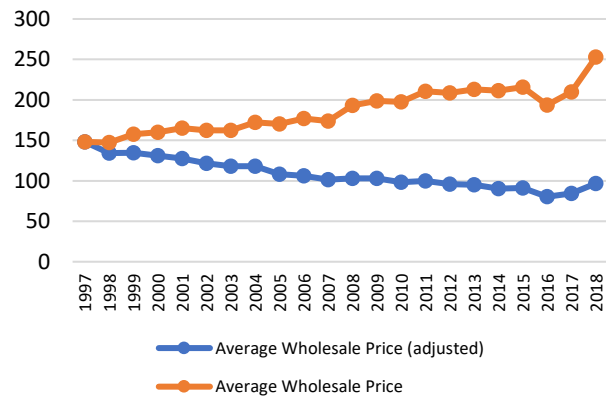
Small dem. fish (< 30 cm)



Dem. fish (> 30 cm)



Int. pelagics



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